

Mapping Cell-Type-Resolved Transcriptional Landscapes of Primary and Recurrent Glioblastoma at Single-Cell Resolution

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Summary

Glioblastoma (GB) remains the most aggressive primary brain tumor in adults and is marked by extensive intratumoral heterogeneity, adaptive cellular plasticity, and near-universal recurrence after standard therapy. Single-cell RNA sequencing (scRNA-seq) has enabled high-resolution dissection of both malignant cell states and tumor microenvironmental populations, revealing that GBM behavior emerges from dynamic interactions between tumor-intrinsic programs, extrinsic immune and stromal signals. Here, a computational scRNA-seq workflow was reconstructed to analyze newly diagnosed (ND) and recurrent (Rec) GB samples. The analysis incorporated adaptive quality control, doublet removal, variance-aware feature selection, batch integration, graph-based clustering, and marker-guided annotation. This framework resolved malignant populations with mesenchymal-like (MES-like), oligodendrocyte progenitor-like (OPC-like), and astrocyte-like (AC-like/Astro-like) features together with tumor-associated macrophages (TAMs), lymphoid cells, and vascular-associated populations. Descriptive compositional comparisons suggested recurrence-associated remodeling of both malignant and immune compartments, consistent with published models of therapy-associated mesenchymal transition, immune evolution, and glioblastoma cell-state plasticity.

Keywords

Glioblastoma, single-cell RNA sequencing, tumor microenvironment, recurrence, mesenchymal state, tumor-associated macrophages, Harmony, Leiden clustering

Introduction

Glioblastoma (GB) is the most common malignant primary brain tumor in adults and remains associated with poor survival despite maximal safe resection, radiotherapy, and temozolomide-based chemotherapy. A central biological challenge in GB is the coexistence of profound intratumoral heterogeneity and adaptive resistance, which together support treatment failure and recurrence. Single-cell transcriptomic studies have reshaped this view by showing that GB is not composed of a uniform

malignant population, but rather of multiple malignant and non-malignant compartments whose relative abundance and functional interactions change over time.

A major conceptual advance came from single-cell analyses demonstrating that malignant glioblastoma cells occupy recurrent transcriptional programs rather than fixed lineages. Neftel et al. defined four major cellular states—mesenchymal-like (MES-like), oligodendrocyte progenitor-like (OPC-like), neural progenitor-like (NPC-like), and astrocyte-like (AC-like)—and showed that these states are linked by substantial plasticity and shaped by both genetics and microenvironmental context. This framework has become central to the interpretation of GB single-cell data because it explains how transcriptionally distinct malignant populations can coexist within the same tumor while remaining capable of interconversion.

Recurrence further complicates this landscape. Single-cell studies of treated glioblastoma have shown that therapy can reshape tumor composition through both cell-intrinsic adaptation and cell-extrinsic remodeling of the tumor microenvironment (TME). Wang et al. reported that glioblastoma evolution under therapy involves coordinated shifts in malignant states and non-malignant compartments, revealing potential therapeutic vulnerabilities in recurrent disease. Similarly, Couturier et al. showed that treatment is associated with an increase in mesenchymal cancer cells and that this transition is linked to immune-dependent changes, supporting the idea that recurrence emerges from ecosystem-level remodeling rather than from malignant cells alone.

The immune compartment is a particularly important contributor to recurrent disease biology. GB is typically enriched in myeloid populations, especially tumor-associated macrophages and microglia, which can reinforce immune suppression, tumor growth, and resistance to therapy. Yeo et al. demonstrated that the immune landscape evolves during glioblastoma progression, with recurrent tumors showing substantial changes in immune composition and signaling. More recent work has further linked macrophage-associated immunoregulatory pathways to impaired anti-tumor immunity in GBM, including checkpoint-like macrophage functions that may restrict productive T cell responses. Together, these studies support the view that recurrence is accompanied by a shift toward a more immunologically remodeled and, in many cases, more mesenchymal tumor ecosystem.

Single-cell analysis is therefore well suited to study recurrence-associated remodeling in GB because it enables simultaneous characterization of malignant plasticity and microenvironmental composition. In this study, a computational workflow was reconstructed to analyze ND and Rec GBM scRNA-seq samples using quality control, doublet detection, integration, clustering, and annotation steps aligned with current best practices. The goal was not to infer causality, but to generate a biologically grounded and methodologically transparent characterization of the cellular architecture of recurrent GB and to interpret these observations in the context of current literature.

Results

Adaptive quality control preserved heterogeneous cell populations while excluding technical outliers

The initial dataset comprised 49,360 cells and 33,694 genes from 12 glioblastoma samples spanning newly diagnosed and recurrent conditions. Exploratory inspection revealed broad inter-patient variability in total UMI counts, detected genes, and mitochondrial transcript fractions, consistent with the heterogeneous quality and composition expected in clinical single-cell datasets (Fig.1). Because rigid filtering thresholds can inadvertently remove biologically relevant low-RNA populations, especially immune cells or stressed malignant states, adaptive filtering based on median absolute deviation (MAD) was applied to quality-control metrics.

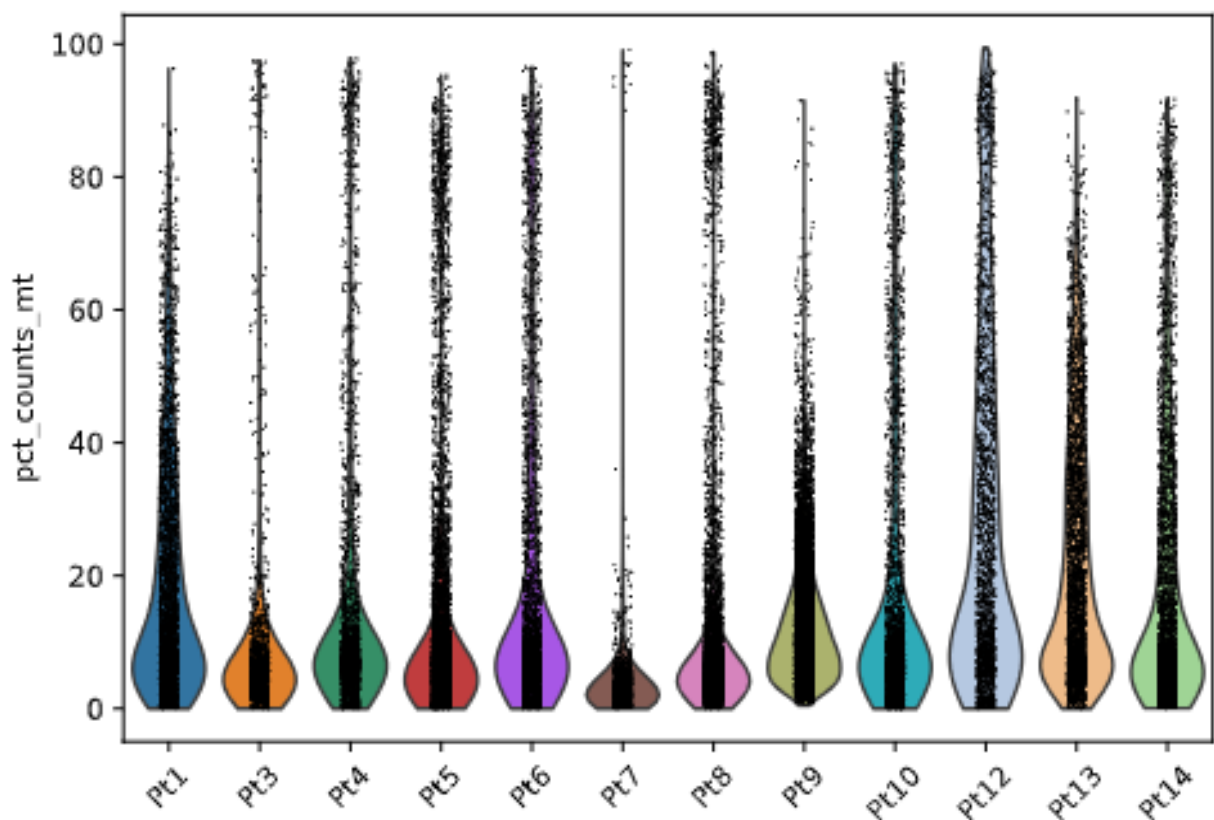


Figure 1. Quality-control metrics across patients. Violin plots show the distribution of total UMI counts, detected genes per cell, and mitochondrial transcript percentages for each patient sample before filtering. Colors or grouped panels represent individual patient datasets. The figure highlights marked inter-sample variability in sequencing depth and RNA complexity, supporting the use of adaptive quality-control thresholds rather than rigid universal cutoffs.

This strategy was chosen because robust outlier detection better accommodates heterogeneous datasets than universal hard cutoffs, a principle emphasized in single-cell best-practice guidelines and quality-

control frameworks. Cells exceeding 5 MADs from the median across selected QC metrics were removed, and genes detected in fewer than three cells were excluded. Following filtering, 37,066 high-quality cells were retained. This step reduced likely debris, damaged cells, and extreme outliers while preserving the broad transcriptional diversity needed for downstream state and cell-type discovery.

Doublet detection reduced artifactual hybrid transcriptional profiles

Doublet detection was performed independently for each patient using Scrublet. This step is particularly important in glioblastoma datasets because the coexistence of malignant, immune, and stromal populations increases the risk that co-encapsulation artifacts will generate hybrid expression profiles that appear biologically meaningful but are technically spurious.

A cluster characterized by elevated Scrublet scores together with incompatible lineage markers, including plasma-cell and myeloid features, was identified and removed before downstream interpretation. This decision was methodologically justified because doublets can distort neighborhood graphs, create artificial clusters, and confound manual annotation (Fig.2). The removal of likely technical hybrids improved interpretability of the integrated embedding and reduced the risk of overcalling rare transitional populations (Fig.5).

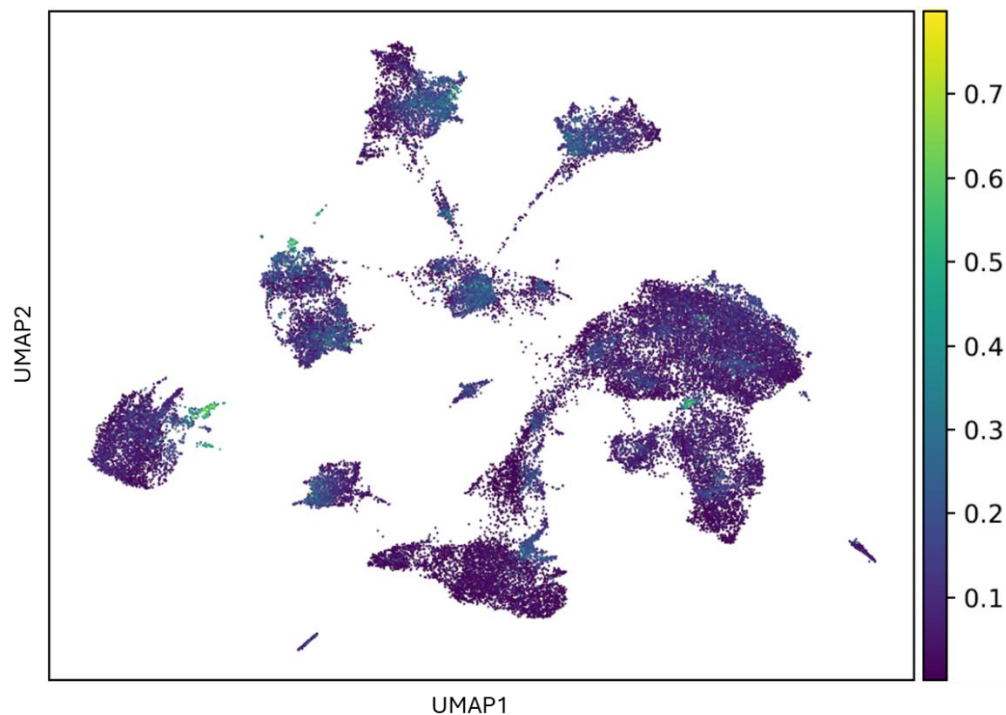


Figure 2. Doublet detection identifies a probable hybrid technical cluster. UMAP embedding of single cells is colored by Scrublet doublet score, with warmer colors indicating higher predicted doublet probability. A discrete cluster

with uniformly elevated scores and mixed lineage-marker expression is consistent with technical co-encapsulation and was excluded from downstream biological interpretation

Variance-aware feature selection and dimensionality reduction captured major transcriptional structure

Raw counts were normalized to a fixed library size and log-transformed before selection of highly variable genes using analytic Pearson residuals. Pearson residual-based feature selection has been recommended for sparse count matrices because it provides improved variance stabilization and reduces the dependence of feature selection on mean expression alone, thereby increasing sensitivity to biologically informative variation in heterogeneous datasets.

Principal component analysis (PCA) on the highly variable gene matrix showed that the first principal components captured the dominant axes of transcriptomic variation, with the elbow plot supporting retention of 30 components for downstream analysis (Fig.3). This dimensionality reduction step compressed correlated gene-expression patterns into a tractable representation while preserving major biological structure needed for integration and graph construction.

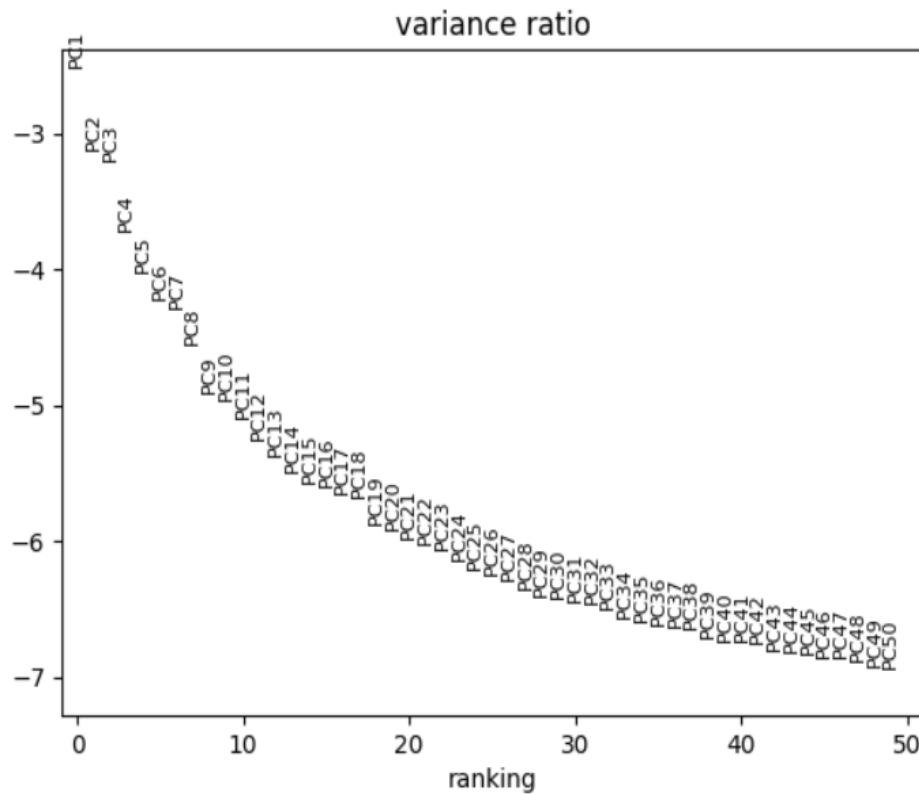


Figure 3. Principal component variance profile supports downstream dimensionality reduction. The elbow plot displays the proportion of variance explained by successive principal components computed from the highly

variable gene matrix. The inflection of the curve near PC30 supports retention of the first 30 components for downstream integration, graph construction, and clustering.

Harmony integration improved cross-patient alignment in a clinically heterogeneous cohort

Patient identity represented a strong source of variation prior to integration, which is expected in clinical GB datasets that contain differences in cellular composition, sequencing depth, and patient-specific biology. Harmony was therefore applied to the PCA embedding using patient identity as a batch variable. Harmony was selected because it is widely used for integration of heterogeneous single-cell datasets and is designed to reduce batch-driven separation while preserving shared cellular manifolds.

Following integration, cells from multiple patients occupied overlapping transcriptional neighborhoods, indicating improved alignment across samples and clinical conditions. This enabled joint clustering and annotation of ND and Rec cells in a common latent space. As with all integration methods, this step requires cautious interpretation because aggressive correction can partially obscure real patient-specific biology; nevertheless, the integrated embedding appeared to preserve coherent lineage structure without obvious visual overcorrection (Fig. 4).

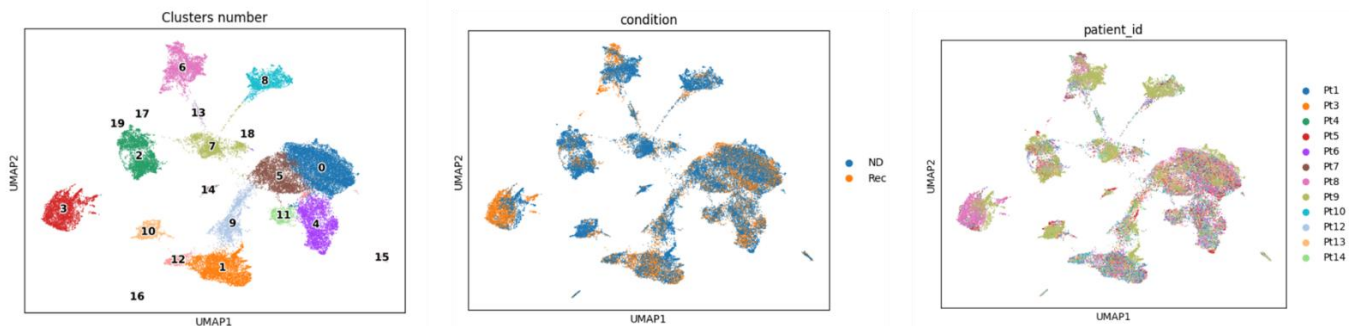


Figure 4. Harmony integration reduces patient-driven separation while preserving shared cellular structure. UMAP projections of the integrated dataset are colored by patient identity and clinical condition. Overlap among cells from multiple patients indicates mitigation of patient-specific batch effects, while preservation of structured manifolds suggests retention of biologically coherent populations after integration.

Graph-based clustering resolved malignant and microenvironmental compartments

A nearest-neighbor graph computed from the Harmony-corrected representation was partitioned using the Leiden algorithm, and low-dimensional visualization was generated with UMAP. Leiden was chosen because it improves partition quality and cluster connectivity relative to earlier graph-partitioning

approaches, which is particularly useful when resolving transcriptionally adjacent populations in large single-cell datasets.

Marker-gene analysis identified transcriptionally coherent clusters corresponding to malignant states and tumor microenvironmental compartments. Manual annotation supported the presence of MES-like, OPC-like, and Astro-like malignant populations together with TAM-rich myeloid populations, T cells, B cells, NK/cytotoxic lymphoid populations, endothelial cells, activated endothelial cells, plasma cells, proliferative populations, and vascular-associated smooth-muscle/pericyte-like cells. The coexistence of these populations is consistent with glioblastoma as a multicellular ecosystem rather than a purely tumor-cell-autonomous entity.

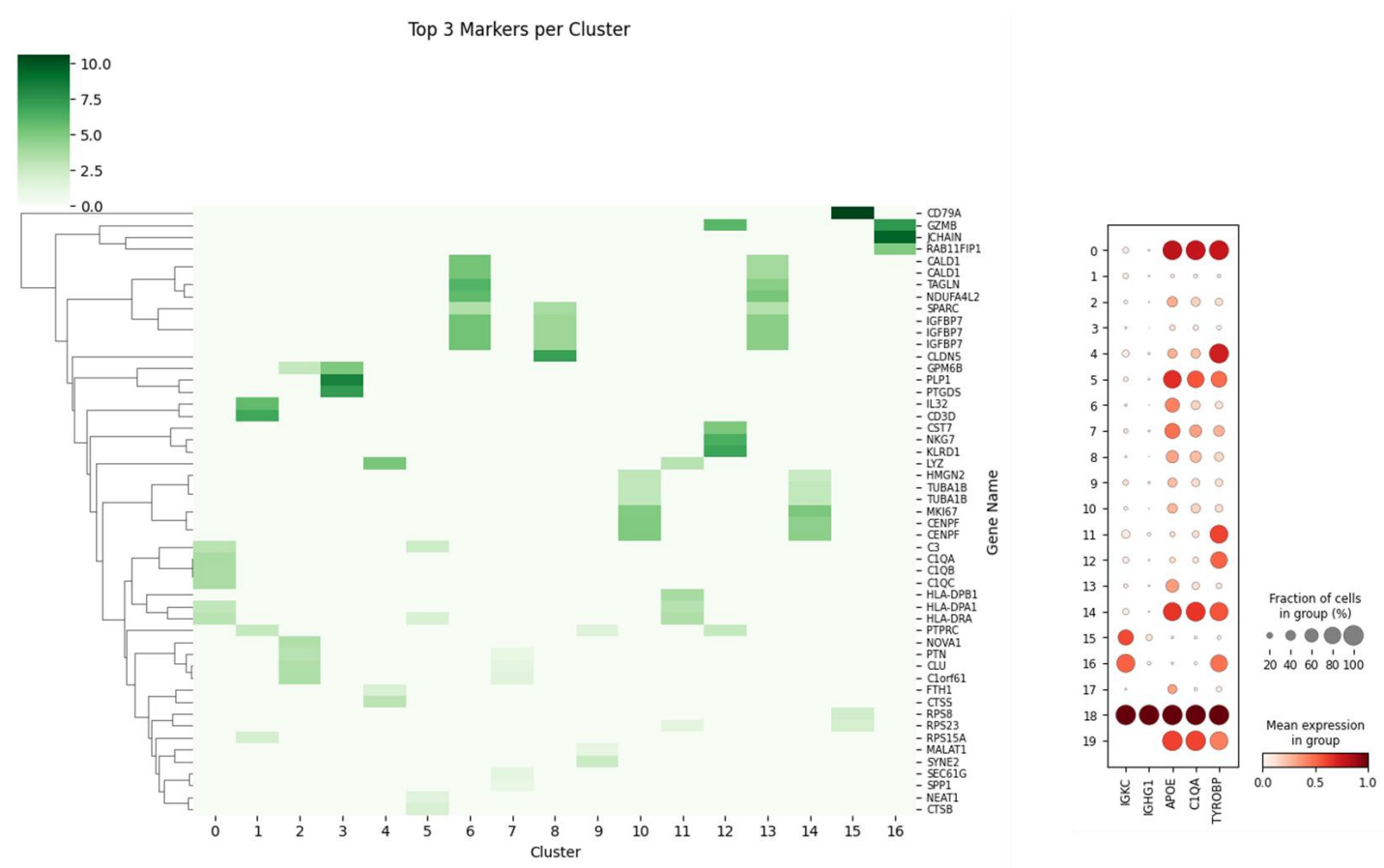


Figure 5. Cluster-resolved marker expression supports annotation of malignant and microenvironmental populations. Dotplot and heatmap summarize representative marker-gene expression across Leiden clusters. Dot size represents the fraction of cells expressing each gene, and color intensity reflects scaled average expression. The marker patterns support annotation of MES-like, OPC-like, and Astro-like malignant states together with immune and vascular-associated compartments.

Malignant-state annotation was consistent with published GB plasticity models

The malignant compartments recovered in the dataset aligned with established single-cell models of glioblastoma cell states. The MES-like population is notable because mesenchymal programs in GB have repeatedly been associated with inflammatory signaling, stress adaptation, and recurrence-associated remodeling. In the current dataset, the identification of a MES-like compartment is therefore biologically compatible with published reports that recurrent and treated tumors often show increased mesenchymal features, although the present analysis remains descriptive and does not establish directional state transitions.

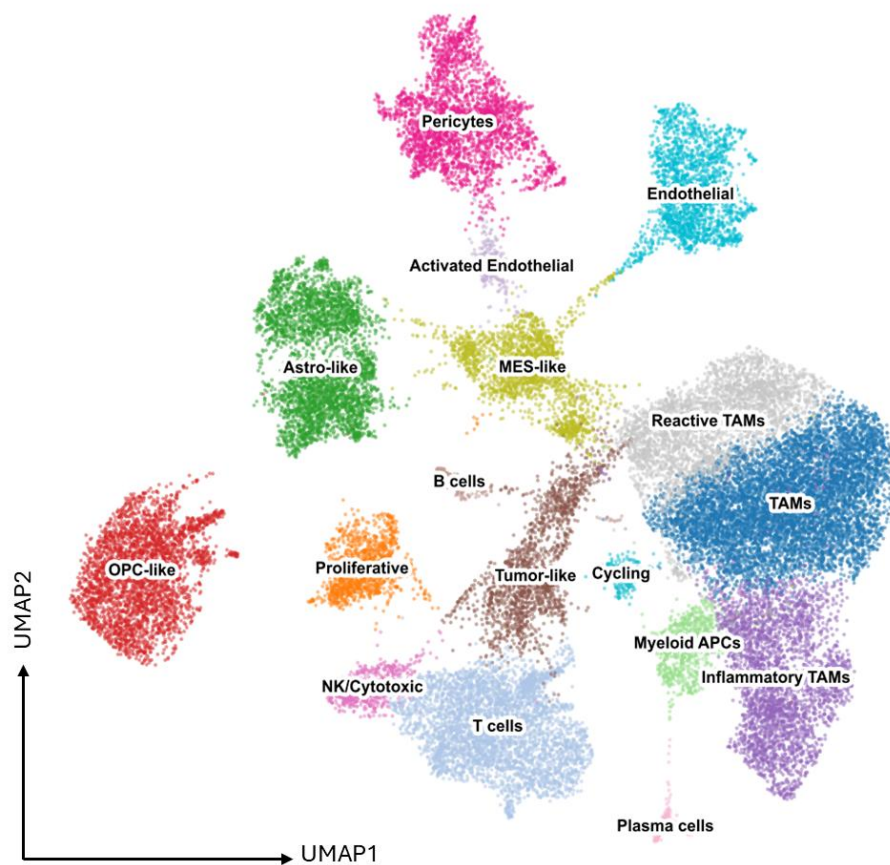


Figure 6. Annotated UMAP reveals the cellular ecosystem of glioblastoma. UMAP embedding of the integrated single-cell dataset is colored by manually curated cell-type or cell-state labels. Distinct transcriptional neighborhoods correspond to malignant, myeloid, lymphoid, and vascular-associated populations, illustrating the multicellular organization of the glioblastoma tumor microenvironment.

The OPC-like and Astro-like populations further support the presence of developmental-lineage-associated transcriptional diversity within malignant cells. In the framework proposed by Neftel et al.,

OPC-like and AC-like states represent distinct but plastic malignant programs rather than stable terminal identities. Their simultaneous presence in this analysis is therefore consistent with cell-state coexistence and transcriptional flexibility in GBM. Recent preclinical single-nucleus studies have also reinforced the relevance of these state-based interpretations for modeling common glioblastoma subtypes (Fig.6).

The immune microenvironment was dominated by TAM-like populations and showed features compatible with immunosuppressive remodeling

A prominent myeloid compartment was observed, including TAM-like clusters expressing canonical macrophage-associated markers. This finding is concordant with prior literature showing that glioblastoma commonly contains abundant tumor-associated macrophages and microglia, which shape cytokine signaling, antigen presentation, matrix remodeling, and immunoregulatory interactions within the tumor microenvironment (Fig.7).

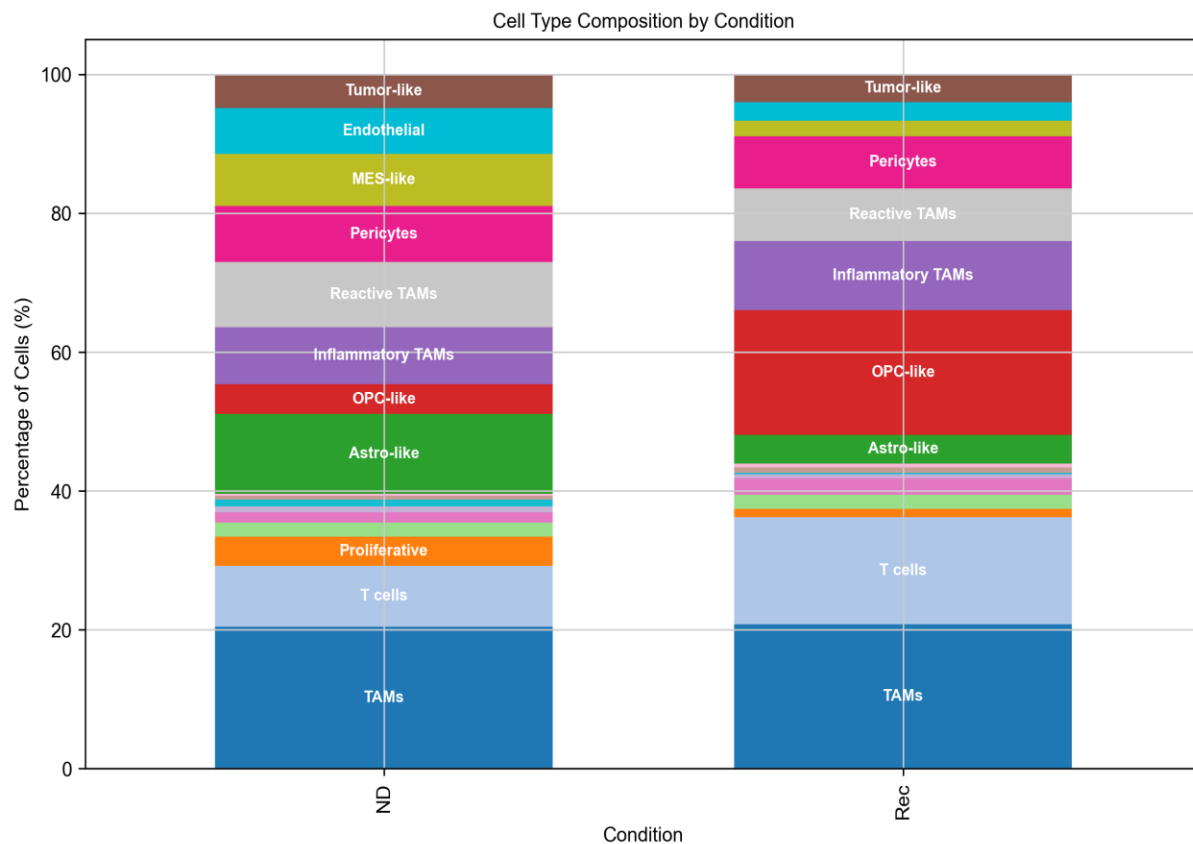


Figure 7. Descriptive compositional comparison suggests remodeling between newly diagnosed and recurrent glioblastoma. Stacked bar plots show the relative proportion of each annotated cell population stratified by clinical condition, with colors representing cell-state or cell-type identities. The figure suggests shifts in malignant and

myeloid compartments between ND and Rec groups; however, these differences are descriptive only and were not evaluated with formal differential abundance statistics.

The prominence of TAM-like populations is biologically relevant because macrophage-enriched GBM ecosystems have been linked to suppression of effective anti-tumor immunity and to support of mesenchymal or treatment-adapted tumor programs. Work by Yeo et al. and Mei et al. supports the interpretation that recurrent GBM immune landscapes can evolve toward stronger immunoregulatory circuitry, including macrophage-associated restriction of T cell priming. Accordingly, the current dataset is consistent with an immune environment in which myeloid populations may contribute to recurrence-associated remodeling, although direct functional immunosuppression was not tested here.

Descriptive compositional analysis suggested recurrence-associated ecosystem remodeling

Stacked-bar visualizations of annotated populations suggested visible differences in the relative abundance of malignant and immune compartments between ND and Rec conditions. In particular, the recurrent setting appeared to show shifts affecting myeloid-rich and malignant-state compartments, consistent with the idea that recurrence reflects coordinated remodeling of tumor-intrinsic and tumor-extrinsic programs.

These observations should be interpreted cautiously. The current analysis did not implement formal differential abundance testing, longitudinal trajectory modeling, or matched-patient evolutionary inference. Therefore, the results support only a descriptive conclusion: ND and Rec samples differed in apparent cellular composition in a manner that is broadly consistent with published reports of therapy-associated mesenchymal enrichment, immune evolution, and recurrent ecosystem restructuring in glioblastoma.

Discussion

This study reconstructed a single-cell RNA-seq workflow to examine cellular organization in newly diagnosed and recurrent glioblastoma and identified a multicompartment ecosystem composed of malignant cell states, immune populations, and vascular-associated cells. The overall pattern is highly consistent with current conceptual models of GBM, in which malignant plasticity and microenvironmental context jointly shape disease progression. Rather than supporting a single dominant lineage, the data fit a state-based framework in which MES-like, OPC-like, and Astro-like programs coexist within the same disease space.

The MES-like compartment is particularly notable in the context of recurrence. Prior studies have shown that mesenchymal transition in GBM is associated with therapy exposure, inflammatory

signaling, and interactions with the immune microenvironment. Couturier et al. reported a treatment-induced and immune-dependent increase in mesenchymal cancer cells, while Wang et al. showed that therapy reshapes both intrinsic malignant programs and extrinsic cellular interactions. Within that context, the apparent MES-like representation observed here is compatible with recurrence-associated adaptation, although the current analysis cannot determine whether mesenchymal enrichment reflects lineage conversion, selective survival, or sampling composition.

The identified OPC-like and Astro-like malignant populations also reinforce the view that recurrent glioblastoma remains developmentally heterogeneous. In the Neftel model, these states are part of a plastic transcriptional continuum shaped by both oncogenic programs and microenvironmental influences. Their presence alongside MES-like cells suggests that recurrence does not eliminate heterogeneity; rather, recurrence may occur through redistribution within an already flexible state landscape. This interpretation is also compatible with recent glioblastoma single-cell and single-nucleus studies that continue to recover related state architectures across experimental and clinical settings.

The immune findings further support a recurrence-associated ecosystem model. TAM-rich myeloid populations were a major component of the dataset, in line with extensive literature showing that myeloid cells dominate the GBM immune microenvironment. Yeo et al. demonstrated that immune composition and signaling evolve during glioblastoma progression, and Mei et al. further highlighted macrophage-centered immune-checkpoint activity that can restrict productive T cell priming. The current observations are therefore consistent with a microenvironment in which myeloid enrichment and immune regulation accompany recurrent disease, even though the present data do not directly measure immune-cell function.

Methodologically, the workflow aligns well with current best practices for single-cell analysis. Adaptive MAD-based filtering reduced reliance on fixed thresholds, Scrublet minimized artifactual hybrid profiles, Pearson residuals improved variance-aware feature selection, Harmony supported integration of patient-derived samples, and Leiden clustering facilitated stable graph partitioning. Collectively, these choices strengthened interpretability in a highly heterogeneous clinical dataset. At the same time, each step involves assumptions that can influence downstream biological conclusions, especially when studying compositional changes across patients and conditions.

Limitations of the study

Several limitations should temper interpretation of these findings. First, the compositional analyses were descriptive and did not include formal differential abundance modeling; thus, observed ND-versus-Rec shifts should not be treated as statistically validated changes. Second, limited clinical metadata restricted deeper interpretation of treatment history, molecular subtype, sampling location,

survival association, or patient matching status. Third, cell-state and cell-type annotations were based on marker expression and literature-guided inference, which may be less reliable for transitional or hybrid populations. Fourth, batch integration with Harmony, while useful for cross-patient comparison, may partially compress biologically meaningful inter-patient structure. Finally, without longitudinally paired samples and dedicated trajectory or clonal analyses, the study cannot directly resolve evolutionary paths leading from primary to recurrent disease.

STAR Methods

Resource availability

Materials availability

This study did not generate new unique reagents.

Data and code availability

- This paper analyzes an existing single-cell RNA-seq dataset derived from 12 glioblastoma samples, including newly diagnosed and recurrent conditions.
- The present report reconstructs and interprets the computational workflow described in the source materials. Public accession identifiers and repository links should be inserted here if available in the final project records.
- This manuscript does not report newly deposited original code.
- Any additional information required to reanalyze the workflow described here should be available from the lead contact upon request.

Experimental model and study participant details

The analysis was performed on previously generated human glioblastoma single-cell transcriptomic data comprising 12 patient-derived samples spanning newly diagnosed and recurrent disease states. No new human enrollment, tissue collection, or experimental intervention was performed for the present manuscript revision. Because the currently available project materials do not provide full demographic and clinicopathologic metadata, variables such as age, sex, molecular subtype, anatomical region, treatment history, and sample pairing status could not be systematically incorporated into the present interpretation and should be explicitly added if available for final submission.

Method details

Data loading and initial inspection

Raw single-cell count matrices were imported into AnnData objects and analyzed using the Scanpy/scverse ecosystem. AnnData provides an efficient annotated matrix structure for single-cell data, while Scanpy offers scalable implementations of preprocessing, dimensionality reduction, graph construction, and marker analysis suitable for datasets of this size. Initial inspection included dataset dimensions, patient labels, and clinical-condition metadata to assess cohort composition before preprocessing.

Quality control and adaptive filtering

Per-cell quality-control metrics were calculated using standard single-cell preprocessing functions, including total UMI counts, number of detected genes, and mitochondrial transcript fraction. Quality control was based on adaptive outlier detection using MAD rather than fixed thresholds. This choice was motivated by current best-practice recommendations emphasizing that rigid universal cutoffs can remove biologically valid cells when datasets contain heterogeneous cell types, variable RNA content, or clinical sample-to-sample differences. Genes detected in fewer than three cells were excluded, and cells exceeding 5 MADs from the dataset-specific median for selected QC metrics were removed.

Doublet detection

Doublets were identified with Scrublet on a per-patient basis before final annotation. Scrublet was chosen because it is a standard computational approach for identifying putative cell doublets directly from expression profiles in droplet-based single-cell datasets. Doublet-aware preprocessing is especially important in glioblastoma because mixed malignant-immune transcriptomes can mimic rare cell states or false transitional populations. Clusters with high doublet scores and incompatible marker combinations were excluded from downstream biological interpretation.

Normalization and selection of highly variable genes

Library-size normalization was performed to a target total of 10,000 counts per cell, followed by log transformation. Highly variable genes were selected using analytic Pearson residuals. This approach was chosen because residual-based variance modeling can better recover informative genes in sparse count data than methods based only on mean-variance heuristics, and it is increasingly recommended in modern single-cell workflows for heterogeneous datasets.

Dimensionality reduction and batch integration

PCA was applied to the highly variable gene matrix to summarize dominant transcriptional structure in a reduced latent space. The first 30 principal components were retained based on the variance-explained profile. Because inter-patient variability can dominate embeddings in clinical scRNA-seq datasets, Harmony integration was then performed using patient identity as the batch covariate. Harmony was selected because it provides efficient correction of batch-associated structure while generally preserving shared biological manifolds across heterogeneous datasets.

Graph construction, clustering, and visualization

A neighborhood graph was computed from the Harmony-corrected representation using 15 neighbors. Leiden clustering was applied at a resolution of 0.5 to identify communities of transcriptionally similar cells. Leiden was preferred because it produces well-connected partitions and improved stability compared with earlier graph-partitioning methods. UMAP was used for low-dimensional visualization of the integrated graph structure.

Marker analysis and cell-state annotation

Cluster markers were identified using the Wilcoxon rank-sum test as implemented in Scanpy. Clusters were then annotated manually by combining marker-gene enrichment with prior glioblastoma single-cell literature. Broad categories included malignant states, myeloid populations, lymphoid populations, endothelial compartments, and vascular-associated cells. Malignant-state interpretation was guided primarily by the GBM state framework described by Neftel et al., while microenvironmental interpretation was informed by studies of recurrent immune remodeling and macrophage-centered immunoregulation in glioblastoma.

Compositional analysis

Relative proportions of annotated cell populations were summarized by clinical condition using stacked bar plots. These analyses were intended as descriptive visualizations of ecosystem composition rather than formal inferential tests. Because no dedicated differential abundance model was applied, apparent differences between ND and Rec groups were not interpreted as statistically confirmed changes.

Quantification and statistical analysis

This study primarily reports computational preprocessing, dimensionality reduction, clustering, annotation, and descriptive compositional analysis. Marker-gene ranking was performed using the Wilcoxon rank-sum framework implemented in Scanpy. No formal hypothesis-testing framework for differential abundance, no longitudinal evolutionary model, and no causal inference strategy were

applied. Therefore, interpretation is restricted to descriptive structure in the analyzed dataset and literature-supported contextualization of the recovered cellular populations.

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